9. Unitarity and reducibility

Outline only — this chapter is not yet written up. The section headings below show the material to be covered.

9.1. Unitary representations and the Haar measure

9.2. The regular representation

9.3. Reducible and irreducible representations

Examples

Example 9.1 (One-dimensional reps of $\mathbb{Z}_2$). Let $G = \mathbb{Z}_2 = \{1, -1\}$ and $V = \mathbb{R}$. The trivial representation is

$$\rho_+(1) = \rho_+(-1) = 1.$$

The sign representation is

$$\rho_-(1) = 1, \qquad \rho_-(-1) = -1.$$

Example 9.2 (Two-dimensional rep of $\mathbb{Z}_2 \cong S_2$). Let $G = \mathbb{Z}_2 \cong S_2 = \{e, \tau\}$ and let $V = \mathbb{R}^2$. Define

$$M(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad M(\tau) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We diagonalize $M(\tau)$ using

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad A^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix},$$

so that

$$\widetilde{M}(\tau) = A^{-1} M(\tau) A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In this basis the representation splits as a direct sum

$$(V, \rho) \cong \rho_+ \oplus \rho_-,$$

so the representation is completely reducible.

Example 9.3 (Irreducible characters of $U(1)$). Let $G = U(1) = \{z \in \mathbb{C} : |z| = 1\}$ and $V = \mathbb{C}$. For any integer $n \in \mathbb{Z}$, define

$$\rho_n(z) = z^n.$$

Then

$$\rho_n(z_1 z_2) = (z_1 z_2)^n = z_1^n z_2^n = \rho_n(z_1)\,\rho_n(z_2),$$

so ρ_n is a one-dimensional representation of $U(1)$. Why n has to be an integer? Write $z = e^{i\theta}$. Any continuous one-dimensional representation must be of the form

$$\rho(z) = e^{ik\theta}$$

for some $k \in \mathbb{R}$. For ρ to be single-valued on the circle we need

$$e^{ik(\theta+2\pi)} = e^{ik\theta} \quad \Longrightarrow \quad e^{2\pi i k} = 1,$$

hence $k \in \mathbb{Z}$. Therefore the irreducible characters of $U(1)$ are precisely

$$\rho_n(z) = z^n, \qquad n \in \mathbb{Z}.$$

Thus there are no other irreducible representations of $U(1)$ beyond these characters.

Complete reducibility for abelian groups

Proposition 9.4. *Every finite-dimensional unitary representation of an abelian group over $\mathbb{C}$ is completely reducible into one-dimensional subrepresentations.*

Sketch of Proof. Let (V, ρ) be a finite-dimensional unitary representation of an abelian group G over $\mathbb{C}$. Choose an orthonormal basis such that all $M(g)$ $(g \in G)$ are unitary matrices:

$$M(g_1)M(g_2) = M(g_2)M(g_1) \qquad \forall\, g_1, g_2 \in G,$$

so the family $\{M(g)\}_{g \in G}$ is a commuting family of normal matrices. By the spectral theorem, these can be simultaneously diagonalized: there is a basis of V in which

$$M(g) = \mathrm{diag}\{\lambda_1(g), \lambda_2(g), \ldots, \lambda_d(g)\}$$

for all $g \in G$. Each coordinate subspace corresponding to one diagonal entry is one-dimensional and G-invariant, so V decomposes as a direct sum of one-dimensional subrepresentations. □

For $G = U(1)$, any finite-dimensional representation on $V \cong \mathbb{C}^d$ therefore has the form

$$M(z) = \mathrm{diag}\big(\rho_{n_1}(z), \rho_{n_2}(z), \ldots, \rho_{n_d}(z)\big)$$

for some integers $n_1, \ldots, n_d$. Equivalently,

$$V \cong \rho_{n_1} \oplus \rho_{n_2} \oplus \cdots \oplus \rho_{n_d}.$$

Domain dependence: real vs complex representations

Consider the standard representation of $SO(2)$ on $\mathbb{R}^2$:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Over $\mathbb{C}$, after a change of basis, we can write

$$R(\theta) \sim \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix},$$

so the complexified representation splits as

$$\rho_{+1} \oplus \rho_{-1},$$

and is thus reducible over $\mathbb{C}$. However, over $\mathbb{R}$ there is no nontrivial 1-dimensional invariant subspace for this action of $SO(2)$, so the representation on $\mathbb{R}^2$ is irreducible. This illustrates that reducibility can depend on the base field.

Unitary representations and complete reducibility

The Euclidean group as a matrix group

Consider the (orientation-preserving) Euclidean group in three dimensions

$$E(3) = \mathbb{R}^3 \rtimes SO(3).$$

An element is written as a pair $(R, \vec{t})$ with $R \in SO(3)$ a rotation and $\vec{t} \in \mathbb{R}^3$ a translation, acting on $\vec{r} \in \mathbb{R}^3$ by

$$(R, \vec{t}): \ \vec{r} \longmapsto R\vec{r} + \vec{t}.$$

The group law is

$$(R_1, \vec{t}_1)\,(R_2, \vec{t}_2) = (R_1 R_2,\ R_1\vec{t}_2 + \vec{t}_1),$$

which is the usual semidirect product law. We can realise this as a matrix representation on $\mathbb{R}^4$ via

$$(R, \vec{t}) \longmapsto \begin{pmatrix} R & \vec{t} \\ 0 & 1 \end{pmatrix},$$

so that

$$\begin{pmatrix} R_1 & \vec{t}_1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} R_2 & \vec{t}_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} R_1 R_2 & R_1\vec{t}_2 + \vec{t}_1 \\ 0 & 1 \end{pmatrix}.$$

The 3-dimensional subspace corresponding to the rotational part is invariant.

Orthogonal complements and complete reducibility

Proposition 9.5. *Let (T, V) be a unitary representation of a group G on a complex (or real) inner product space V, and let $W \subset V$ be an invariant subspace. Then the orthogonal complement*

$$W^{\perp} = \{y \in V \mid \langle y, x \rangle = 0 \ \forall x \in W\}$$

is also G-invariant.

Proof. Let $g \in G$ and $y \in W^{\perp}$. We must show that $T(g)y \in W^{\perp}$. For any $x \in W$,

$$\langle T(g)y, x \rangle = \langle y, T(g)^{\dagger} x \rangle = \langle y, T(g^{-1})x \rangle,$$

using unitarity. Since W is invariant, $T(g^{-1})x \in W$, and $y \perp W$ implies $\langle y, T(g^{-1})x \rangle = 0$. Hence $\langle T(g)y, x \rangle = 0$ for all $x \in W$, so $T(g)y \in W^{\perp}$ and $W^{\perp}$ is invariant. □

Corollaries

1. *Finite-dimensional unitary representations are completely reducible.* If V is reducible, we have $V \cong W \oplus W^{\perp}$ with both subspaces invariant. If either W or $W^{\perp}$ is still reducible, we repeat the construction. In finite dimension this process terminates after finitely many steps, so

   $$V \cong W_1 \oplus W_2 \oplus \cdots \oplus W_k$$

   with each W_j irreducible. The finite-dimensional hypothesis is essential: for instance, the (one-dimensional) unitary irreps of $\mathbb{R}$ are

   $$\rho_k(a) = e^{ika}, \qquad k \in \mathbb{R},$$

   and the regular representation on $L^2(\mathbb{R})$ decomposes into a continuum of such characters.

2. *For compact groups, every finite-dimensional representation is unitarizable and hence completely reducible.* Given a representation (ρ, V) of a compact group G, average any inner product over G using the Haar measure to obtain a G-invariant inner product. Thus ρ is equivalent to a unitary representation, and by the previous item it is completely reducible.

3. *Finite groups and the regular representation.* For a finite group G, consider the Hilbert space $L^2(G) \cong \mathbb{C}^{|G|}$ with orthonormal basis $\{\delta_g\}_{g \in G}$, where $\delta_g(h) = \delta_{gh}$. The left regular representation is

   $$(L_h \delta_g)(k) = \delta_g(h^{-1}k) = \delta_{hg}(k).$$

   This is a unitary representation of dimension $|G|$, hence completely reducible. We will use $G = S_3$ as a concrete example below.

Examples not completely reducible

The previous examples were completely reducible; now consider the following.

Example 9.6. Let
$$U(x) = \begin{pmatrix} x & 1 \\ 0 & 1 \end{pmatrix}, \qquad x \in K,$$
acting on K^2 ($K = \mathbb{R}$ or $\mathbb{C}$). The subspace
$$W = \operatorname{span}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
is invariant, but there is no invariant complement, so this representation is not completely reducible.

Example 9.7. Let $A \in GL(n, K)$ and consider the map
$$A \longmapsto \log\left|\det A\right|,$$
which is a homomorphism from $GL(n, K)$ to $(\mathbb{R}, +)$:
$$\log|\det(AB)| = \log|\det A| + \log|\det B|.$$
This gives a representation on K^2 by
$$T(A) = \begin{pmatrix} 1 & \log|\det A| \\ 0 & 1 \end{pmatrix},$$
since
$$T(A)T(B) = \begin{pmatrix} 1 & \log|\det A| + \log|\det B| \\ 0 & 1 \end{pmatrix} = T(AB).$$
Again the line span $\binom{1}{0}$ is invariant but has no invariant complement, so the representation is not completely reducible.

Semidirect products

Let H and G be groups. Recall that the direct product $H \times G$ has multiplication
$$(h_1, g_1)(h_2, g_2) = (h_1 h_2, g_1 g_2).$$
A *semidirect product* $H \rtimes_\alpha G$ uses in addition an action $\alpha : G \to \operatorname{Aut}(H)$ and is defined by
$$(h_1, g_1)(h_2, g_2) = \big(h_1\, \alpha_{g_1}(h_2),\ g_1 g_2\big).$$
The direct product is the special case where the action is trivial, $\alpha_g = \mathrm{id}_H$. The Euclidean group example above is of the form $\mathbb{R}^3 \rtimes SO(3)$.

Symmetric groups and standard representations

The canonical representation of S_3 on $\mathbb{R}^3$

Consider the natural action of S_3 on $\mathbb{R}^3$ with basis $\{e_1, e_2, e_3\}$:

$$T(\sigma)e_i = e_{\sigma(i)}.$$

Define

$$u_0 = e_1 + e_2 + e_3.$$

Then u_0 spans a one-dimensional invariant subspace $W = \operatorname{span}\{u_0\}$ carrying the trivial representation. A complementary subspace is

$$W^\perp = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\} = \operatorname{span}\{u_1, u_2\},$$

where we may choose

$$u_1 = e_1 - e_2, \qquad u_2 = e_2 - e_3.$$

In this basis one finds explicit 2×2 matrices for the action of the generators of S_3; for example,

$$T\big((12)\big)u_1 = -u_1, \qquad T\big((23)\big)u_1 = u_1 + u_2, \quad \text{etc.}$$

Thus $\mathbb{R}^3 \cong W \oplus W^\perp$, with W the trivial irrep and $W^\perp$ a 2-dimensional irrep (the standard representation). Using an orthonormal basis of $W^\perp$ gives a unitary realisation of this 2-dimensional irrep.

The standard representation of S_n

More generally, let S_n act on $\mathbb{R}^n$ by permuting the standard basis $\{e_1, \dots, e_n\}$,

$$T(\sigma)e_i = e_{\sigma(i)}.$$

The vector

$$u_0 = \sum_{i=1}^n e_i$$

spans a 1-dimensional invariant subspace W (trivial representation). The orthogonal complement

$$W^\perp = \left\{ x = \sum_{i=1}^n x_i e_i \in \mathbb{R}^n \;\middle|\; \sum_{i=1}^n x_i = 0 \right\}$$

is S_n-invariant and is called the *standard representation.*

Proposition 9.8. *For $n \geq 3$, the standard representation $W^\perp$ of S_n is irreducible.*

Sketch. Let $U \subset W^\perp$ be a non-zero invariant subspace, and take $u = \sum_i x_i e_i \in U$ not proportional to u_0; hence not all x_i are equal. After permuting coordinates we may assume $x_1 \neq x_2$. Consider

$$u - (12)u = (x_1 - x_2)(e_1 - e_2) \in U,$$

so $e_1 - e_2 \in U$. Applying arbitrary permutations shows that $e_i - e_j \in U$ for all i, j, hence

$$\operatorname{span}\{e_i - e_j \mid 1 \leq i < j \leq n\} \subset U.$$

This span has dimension $n-1$ and equals $W^\perp$, so $U = W^\perp$. $\square$

Thus the canonical permutation representation of S_n on $\mathbb{R}^n$ decomposes as

$$\mathbb{R}^n \cong \mathbb{1} \oplus W^\perp$$

(trivial plus standard irrep).

Regular representation of S_3 and isotypic decomposition

We now return to the regular representation of S_3 and its decomposition into irreducibles.

Character table and decomposition of $L^2(S_3)$

The group S_3 has three conjugacy classes and hence three irreducible representations. A convenient character table is

χ	e	$3(12)$	$2(123)$
V_1	1	1	1
V_1'	1	-1	1
V_2	2	0	-1

where V_1 is the trivial representation, V_1' the sign representation, and V_2 the 2-dimensional standard irrep. For the regular representation ρ_{reg} on $L^2(S_3)$ we have

$$\chi_{\text{reg}}(e) = |S_3| = 6, \qquad \chi_{\text{reg}}(g) = 0 \quad (g \neq e).$$

We seek integers x, y, z such that

$$\rho_{\text{reg}} \cong xV_1 \oplus yV_1' \oplus zV_2.$$

On the level of characters this reads

$$\chi_{\text{reg}} = x\chi_1 + y\chi_1' + z\chi_2.$$

Solving

$$\begin{cases} x + y + 2z = 6, \\ x - y + 0z = 0, \\ x + y - z = 0, \end{cases}$$

gives $x = 1$, $y = 1$, $z = 2$. Hence

$$L^2(S_3) \cong V_1 \oplus V_1' \oplus 2V_2.$$

This is a concrete example of *isotypic decomposition.*

Isotypic components in general

Assume that the set of irreducible representations (up to isomorphism) of a group G is countable. Choose one representative

$$(T^{(\mu)}, V^{(\mu)})$$

for each isomorphism class, labelled by μ. Then any finite-dimensional completely reducible representation (T, V) admits a decomposition

$$V \cong \bigoplus_{\mu} \bigoplus_{i=1}^{a_\mu} V^{(\mu)},$$

where a_μ is the multiplicity with which $V^{(\mu)}$ appears. The subspace

$$\bigoplus_{i=1}^{a_\mu} V^{(\mu)}$$

is called the *isotypic component* of V corresponding to μ.

For $a_\mu > 0$, we can identify

$$V^{(\mu)} \oplus \cdots \oplus V^{(\mu)} \;\cong\; K^{a_\mu} \otimes V^{(\mu)} \;=:\; a_\mu V^{(\mu)},$$

where $K = \mathbb{R}$ or $\mathbb{C}$.

Multiplicity spaces and matrices

Let $V^{(\mu)}$ have basis $\{u_i\}_{i=1}^{n_\mu}$ and K^{a_μ} have standard basis $\{e_\alpha\}_{\alpha=1}^{a_\mu}$. Then

$$K^{a_\mu} \otimes V^{(\mu)}$$

has basis $\{e_\alpha \otimes u_i\}$. On this space define the representation of G by

$$g \cdot (e_\alpha \otimes u_i) = e_\alpha \otimes T^{(\mu)}(g) u_i = \sum_j e_\alpha \otimes M^{(\mu)}_{g,ji} u_j,$$

where $M^{(\mu)}_{\mathrm{g}}$ is the matrix of $T^{(\mu)}(g)$ in the chosen basis. Thus the action on $K^{a_\mu} \otimes V^{(\mu)}$ is

$$T(g) = \mathbb{1}_{a_\mu} \otimes T^{(\mu)}(g),$$

i.e. G acts trivially on the multiplicity (or degeneracy) space K^{a_μ} and nontrivially only on $V^{(\mu)}$. In matrix language, a vector in the isotypic component can be written as $\psi_{\mu,i,\alpha}$ (irrep label μ, basis index i, degeneracy index α), and the group action is

$$T(g) : \; \psi_{\mu,i,\alpha} \longmapsto \sum_{j=1}^{n_\mu} M^{(\mu)}_{g,ji} \, \psi_{\mu,j,\alpha}.$$

The irrep label μ and the degeneracy index α are unchanged; only the internal index i is rotated by the irrep matrix.